SPERT® Forecaster

Quick Reference Guide

Agile Release Forecasting with Monte Carlo Simulation · forecaster.spertsuite.com

WHAT IS SPERT® FORECASTER?

Sprint velocity	The amount of work a team completes per sprint (in any unit: story points, hours, items). SPERT® Forecaster derives velocity mean and standard deviation from your sprint history, or accepts a subjective estimate.
Remaining backlog	The total work yet to be done at the time of the forecast. Enter the current backlog size in the same unit as your velocity.
Distribution types	Six distributions are available: T-Normal (Truncated Normal), Lognormal, Gamma, Triangular, Bootstrap (history mode, 5+ sprints), and Uniform. Only Lognormal is enabled by default — add others in Settings → Statistical methods to show. Each enabled distribution produces its own set of percentile results.
Percentile results	P50 = 50% chance of finishing by that sprint/date. P80 = 80% confidence. Higher percentiles give later dates with more buffer. Use P80–P90 for stakeholder commitments.
Forecast modes	History — derives the velocity mean and variability from your recorded sprints. Subjective — uses a point estimate plus a coefficient of variation (CV) when history is thin. **The app chooses for you:** with 5 or more included sprints it starts in History, otherwise Subjective. You can switch to History manually from 2 included sprints onward, provided those sprints do not all share an identical velocity — zero variability would imply a certainty the data cannot support, so the app disables the History button in that case.

■ SPERT® Forecaster can import a product exported from SPERT® Story Map, automatically converting releases into milestones with pre-populated backlog sizes and velocity data.

STEP-BY-STEP WORKFLOW

1. **Projects** — Click **Add Project** to reveal the new-project form, or click **Load Sample Project** for a ready-made example with sprint history, milestones, and a productivity adjustment. Load Sample Project is available at any time from the toolbar — not only when your workspace is empty — so you can pull a fresh reference copy alongside your own work. Enter a name, sprint cadence (1–4 weeks), first sprint start date, and unit of measure (e.g., story points, hours). Each project tile shows action icons: per-tile download (exports that project as JSON), clone (copies all data to a new project), edit, and delete.

2. **Sprint History** — Record completed sprints. For each sprint, enter the *Done this sprint* value (velocity). Optionally enter the backlog remaining at sprint end for burn-up chart accuracy. A compact read-only summary of your 3 most recent sprints appears above the Add Sprint form for quick reference. Toggle sprints in/out of the forecast with the Include checkbox to exclude anomalous sprints.

3. **Forecast tab** — Select History or Subjective mode. Enter the remaining backlog. Velocity mean and variability are auto-populated from history. Once sprint cadence and start date are configured, the simulation runs automatically on first valid input — no manual click required. If a prerequisite is missing, a helper message under the Run Forecast button explains what to set on the Sprint History tab.

4. **Adjust inputs** — Optionally: adjust velocity volatility using the Adjust link (history mode), or click *Reset overrides* to clear all manual velocity and variability overrides at once. Model scope growth from the calculated or custom rate, or switch to Subjective mode to enter a point estimate and predictability rating.

5	**Review charts** — The "Your forecast" hero callout shows the true cumulative probability at the displayed date (capped at 99%). Use the scope picker to switch between Entire Project and individual milestones. The Forecast Results table is collapsed by default — click its header to expand. The **Deadline Probability** panel (below the results table) answers the inverse question: given a target date, what is the probability of finishing by then? Scroll to explore: Burn-Up, CDF, and Histogram charts. Use the Custom Percentile selector for any P1–P99 result. Leaving the Forecast tab and coming back no longer discards your results — they are retained per project, so you get the same dates back rather than a fresh simulation.
6	**Share results** — Use the copy-image button on any chart to capture it to the clipboard, or click the report icon to generate a printable HTML report including the results table and selected charts. Use milestones to show per-release forecast dates.

■ Auto-recalculate (Settings) defaults to ON — the simulation runs automatically whenever inputs are valid. For large trial counts or slow devices, disable auto-recalculate and click Run Forecast manually.

FORECAST INPUTS — FIELD GUIDE

Backlog	Remaining work to complete (required). Shown in the project's unit of measure. The helper text shows the backlog value from the last recorded sprint as a reference. If the stored value drifts from the most recent *included* sprint's ending backlog (e.g., after toggling a sprint's Include status), a *Reset to N* link appears next to the field for one-click recovery.
Velocity (mean)	Average sprint throughput. Auto-populated from history; overridable. In Subjective mode, derived from your point estimate — the field is read-only.
Variability	Velocity spread (standard deviation). A help tooltip on the label reads: "Standard deviation — how much velocity varies sprint to sprint." Auto-calculated from history. Use the *Adjust* link to apply a volatility multiplier (0.75×, 1.0×, 1.25×, 1.5×) instead of entering a raw value. In Subjective mode, derived from the selected CV. When any velocity or variability override is active, a *Reset overrides* link clears all three fields and closes the volatility adjuster in one click.
Start Date	The forecast start date — automatically set to the next sprint's start date based on your sprint cadence and history. For a project that has a first sprint start date set but no completed sprints yet, the forecast anchors on that *first sprint's* start date rather than today, so every derived date (results, CDF, burn-up, deadline probability) is computed from the schedule you entered. Read-only; update via the Sprint History tab.
Cadence	Sprint length in weeks (1–4). Set on the Projects tab; displayed here for reference. Affects how sprint counts translate to calendar dates in all results. Sprint cadence and first sprint start date must be configured before the Run Forecast button is enabled.
Scope growth	Optional. Model backlog expansion sprint-over-sprint (if backlog tends to grow each sprint). Use the calculated rate (derived from Sprint History backlog entries) or enter a custom value. Adds the growth rate to remaining backlog each simulated sprint. The *Model scope growth* toggle is remembered per project — switching projects no longer carries one project's setting into another.
Subjective mode	Enter an estimated velocity and select a predictability rating: **Very steady** (CV 0.15) → **Fairly consistent** (0.25) → **Somewhat variable** (0.35) → **Often disrupted** (0.45) → **Quite volatile** (0.55) → **Wildly uncertain** (CV 0.65). Std Dev = velocity × CV. This is the mode the app starts in until 5 sprints are included, and it stays the right choice whenever your history is too short or too uniform to describe future variability. If your estimate is more than 2× or less than 0.5× the historical average, an amber warning appears — verify the estimate is intentional.

FORECAST RESULTS & CHARTS

Forecast Summary	A blue "Your forecast" hero callout shows the *true cumulative probability* at the displayed sprint-end date — typically higher than the selected percentile because sprint quantization means each sprint bucket covers a range of simulated outcomes. The probability shown in the callout is capped at 99%; the CDF and Histogram charts correctly reach 100% as mathematical visualizations and are not subject to this cap. The scope picker lets you switch between Entire Project and individual milestones; both the hero callout and the per-milestone breakdown update immediately. The summary sentence reads *"there is at least an X% chance"* — the "at least" reflects the sprint-quantization gap. A colored-dot list below shows forecast dates for each milestone at the selected percentile. Completed milestones appear in italic. Results and the summary's scope, distribution and percentile selections are retained per project — leave the Forecast tab and return and the same numbers come back, rather than a re-run producing slightly different dates.
Percentile table	Rows = selectable percentiles (P5–P95 toggleable; default is P10, P50, P90); columns = enabled distribution types. Each cell shows sprints required and the corresponding finish date. The Forecast Results section is **collapsed by default** — click its header to expand. The Custom Percentile dropdown spans P10–P95 (optimistic through conservative). In History mode with fewer than 4 included sprints, an amber warning notes that forecast spread may be understated. Your percentile pill selection and both custom-percentile slots are remembered per project; they no longer re-seed from Settings each time you return to the tab.
Deadline Probability	A collapsible panel (collapsed by default) that inverts the forecast question: given a target date, what is the probability of finishing by then? Set a Target Date, choose a Scope (Entire Project or any not-yet-completed milestone), and select a Distribution. The panel shows a per-distribution probability table and a narrative sentence naming the scope, the date, and the sprint at which the probability is evaluated. Probability is measured at the end of the last forecast sprint whose finish date falls on or before your target date (conservative: "by this date" = "by this sprint boundary"). A sprint-boundary footnote always names the exact sprint and date used so you can audit the result. All probabilities cap at 99%. Target date is session-only and clears on project switch.
Milestone results	If milestones are defined, a per-milestone breakdown of forecast dates at the selected percentile appears in the Forecast Summary panel. The milestone dropdown in the Custom Percentile and chart views lets you focus on a specific release.
Burn-Up chart	Plots completed work (solid line), total backlog scope (dashed line), and three forecast fan lines (P50/P80/P90 or configured). Chart legend is ordered: Scope, Done, then forecast lines ascending by percentile. Hover for sprint details.
CDF chart	Cumulative distribution function — probability of finishing within each number of sprints. Shows all enabled distributions as overlapping curves. Hover to read exact probabilities. Milestone reference lines shown when milestones are defined; completed milestones are excluded from the milestone selector dropdown.
Histogram	Frequency distribution of simulated sprint-to-finish counts across all trials. Visualizes the spread of possible outcomes — a narrow peak = low uncertainty; a wide spread = high volatility. Completed milestones are excluded from the milestone selector dropdown.
Copy & Report	Each chart has a copy-image button (camera icon) to capture it to the clipboard. The report icon in the results section opens a popover to generate a printable HTML report including the results table and any expanded charts. All chart captures render correctly in both light and dark mode.

■ The Bootstrap distribution (history mode, 5+ sprints) resamples actual sprint velocities directly — no distributional assumption. It often gives the most realistic spread when you have a reliable history. In History mode with fewer than 5 included sprints, a footnote below the results table shows how many more sprints are needed to unlock Bootstrap.

MILESTONES & PRODUCTIVITY ADJUSTMENTS

Milestones	Named delivery targets (e.g., MVP, Beta, GA). Each milestone has an incremental backlog size — the work allocated to that milestone only. Milestones are cumulative: the forecast chains them sequentially. Drag to reorder. Up to 10 milestones per project. To fix a name, click it directly on the Milestones panel — the cell becomes a text input in place. Use the Pencil button when you also need to change Remaining Work or Color.
Milestone colors	Each milestone gets an auto-assigned color (emerald, blue, amber, purple, red — cycled). The color appears as a reference line on the Burn-Up and CDF charts when "Show on chart" is enabled.
Productivity adjustments	**Productivity Adjustments (Holidays, Breaks, Events)** — Define date ranges where the team operates at reduced capacity. Enter a start date, end date, and productivity factor: **0%** = no work done, **50%** = half velocity, **100%** = full velocity (no effect). Enabled adjustments reduce effective velocity in the simulation for those sprints. Note: because forecasts report sprint finish dates (not intra-sprint completion dates), a small adjustment may not shift the projected end date if work still falls within the same sprint.
Enable / disable	Each productivity adjustment has a toggle. Disabled adjustments are saved but have no effect on the forecast — useful for modeling scenarios without deleting the adjustment.

■ Milestone backlog sizes are incremental, not cumulative. If your total backlog is 500 points split across MVP (200) and GA (300), enter 200 for MVP and 300 for GA — not 200 and 500. SPERT® Forecaster accumulates them automatically.

SPRINT HISTORY TAB

Sprint config	Set sprint cadence (1–4 weeks), first sprint start date, and project start/finish dates here. Sprint start dates are calculated automatically from the first sprint start date and cadence.
Adding sprints	Click Add Sprint. A compact read-only summary of your 3 most recent sprints appears above the form — showing sprint number, date range, velocity, and backlog at end — so you can reference prior values without closing the form. Enter the *Done this sprint* value (velocity) and optional *Backlog at End* (in the project's unit of measure). Sprint finish dates are calculated and shown; optionally override a finish date to extend a sprint (e.g., for a holiday week or spring break).
Custom finish dates	Override the computed finish date for any sprint when the team worked longer than the standard cadence. Overridden dates cascade forward — all subsequent sprint start dates shift accordingly. A visual indicator marks sprints with custom dates. Use Reset to restore the calculated date.
Include toggle	Exclude anomalous sprints (team at 20% due to vacation, major scope change, etc.) from velocity statistics without deleting the record. Excluded sprints still appear in the burn-up chart. If the Remaining Backlog field on the Forecast tab drifts after a toggle, a *Reset to N* link appears next to the field for one-click correction.
Velocity stats	Calculated mean, std dev, coefficient of variation (CV), and count of included sprints are shown above the sprint list. These feed directly into the Forecast tab.
Velocity chart	A bar chart of per-sprint velocity with a mean reference line. Visually identify trends, outliers, and improving or declining team throughput over time.
Scope analysis	If Backlog at End is recorded for multiple sprints, the Scope Analysis section shows scope injection rate (backlog growth per sprint) — the source data for the calculated scope growth option.

DATA, SETTINGS & STORAGE	
Local storage	Default. Data stays in your browser only — never sent to any server. Safe for corporate data. Clearing browser site data permanently deletes all projects. Use Export to back up.
Cloud storage	Optional. Sign in with Google or Microsoft. Data syncs via Firebase across devices. Enables project sharing with team members (owner / editor / viewer roles).
Export / Import	A download icon on each project tile exports that single project as JSON. The **Export All** button in the toolbar exports every project; Settings → Export Projects lets you select a subset. **Import** opens an inline conflict-resolution preview: each conflicting project shows three choices — *Keep existing* (default for ID conflicts), *Add as a copy*, or *Replace existing*. A copy is named *Name - Copy (N)*, matching the Clone button convention. Non-conflicting projects are listed in a green "New projects" block. A result banner confirms counts after the import completes. In cloud mode the Import button stays disabled until your cloud projects have finished loading, so conflict detection always runs against your real workspace.
Import from Story Map	(Projects tab → Import) Import a product exported from SPERT® Story Map. Releases become milestones; sprint velocity and backlog data are mapped automatically. Story Map imports use the same inline conflict-resolution preview as native Forecaster exports.
Export CSV	Download simulation percentile results as a CSV file for use in reports or presentations.
Settings	Configure: trial count (1K–50K; default 10,000), auto-recalculate toggle (defaults to ON — and it now skips the re-run entirely when nothing affecting the numbers has changed), **Statistical methods to show** (enable/disable individual distributions; default is Lognormal only; at least one must remain enabled), default chart font size, default custom percentiles (two slots), default results percentiles, theme (light/dark/system), and Export Attribution (name and ID embedded in exported JSON for traceability).

■ Set a lower trial count (1,000–5,000) for fast exploratory forecasting, then increase to 10,000–50,000 before a stakeholder presentation for smoother, more stable percentile results.

CONNECT AI	
Connect AI button	Pair an AI assistant — Claude, ChatGPT, Copilot, Gemini and others — with the project you have open, so it can read your forecast and explain it. The button sits in the app header, immediately left of the theme toggle, and is visible from every tab. Once paired it reads **AI** with a status dot: **grey** means an assistant is paired but idle, **pulsing blue** means one is actually connected and reading. The button is hidden until you have at least one project, and requires cloud configuration — it does not appear in a local-only build.
Read-only, always	The connection cannot change anything in SPERT® Forecaster. There is no tool an assistant could call to edit your project — the server does not define one — the pairing is created with writes permanently refused, and the server re-checks that before any write could occur. If your assistant concludes something should change, it will tell you what and where, and you make the change yourself.
Consent & Read Mode	The first connection opens a consent dialog. **Read Mode** is the only toggle: with it on, a copy of the *currently open* project is uploaded to SPERT's cloud so the assistant can read it. Only that one project is ever uploaded — never your whole workspace — and switching projects replaces the copy rather than accumulating them. Turning Read Mode off deletes the uploaded copy immediately and keeps the pairing alive; with it off, the assistant can still explain how forecasting works but cannot see your data. Once you have consented, later connections from the same browser resume silently and go straight to the pairing code.

Pairing code & prompt	The panel shows a single-use pairing code that expires in 15 minutes (it refreshes every 10 minutes while the panel is open and waiting). Two buttons copy what you need: **Copy code** for the code alone, and **Copy prompt for your AI** for a prepared prompt that carries the code and tells your assistant how to read the data correctly — use the prompt unless you have a reason not to. Your assistant needs the SPERT MCP server (mcp.spertsuite.com) added as a connector first. A pairing stays valid for 7 days from your last activity.
What the assistant reads	Your project configuration, sprint history and velocity statistics, the inputs the forecast actually ran on, percentile results per distribution and per milestone, your productivity adjustments, and your deadline probability. It does *not* receive the raw per-trial data, the charts as drawn, any other project, the cumulative-probability curve, or the large percentage in the "Your forecast" headline — that figure is a different calculation and reads higher than the percentile below it. The data carries its own list of what it omits, so the assistant can say what it cannot see rather than guessing.
Freshness status	Every read is labelled, so an assistant need not guess whether your numbers are current: **fresh** (the results match your current inputs), **stale** together with the specific reason — a milestone was resized, the included sprint set changed, a productivity adjustment moved — **recomputing** (a run is in progress), or **absent** (no forecast has been run in this browser session). Ask your assistant to check the status before it quotes a date.
Visible vs. computed	The app computes up to six distributions but you may be looking at one. The data says which distributions are actually visible to you as distinct from which were computed, and separates the ones excluded because they are invalid for your current mode from the ones you chose to hide in Settings. Each scope is likewise flagged with whether it is the one on your screen, so an assistant will not quote a milestone you are not reading.
Disconnect	**Disconnect** in the panel ends the pairing and deletes both the session and the uploaded copy. Signing out does the same. Turning Read Mode off deletes the uploaded copy but keeps the pairing, and switching from cloud to local storage does likewise. The panel also closes on **Done**, on Escape, or on a click outside it.

■ Tell your assistant this is a **Monte Carlo simulation over sprint velocity** — it is **not** three-point PERT, and there is no (O + 4M + P) / 6 anywhere in it. Assistants get this wrong reliably and confidently. The copied prompt says so, and two explainer tools (available to any assistant connected to the SPERT MCP server, with or without a pairing) cover the forecasting method and this app's vocabulary. Ask your assistant to consult them before it explains the method.

■ Two things worth knowing. **Read Mode uploads regardless of your storage setting** — if you work in Local storage, enabling Read Mode still sends a copy of the open project to SPERT's cloud for as long as the pairing lasts. And a pairing left open for days with no edits can lose its uploaded copy to expiry while still looking connected; if your assistant reports no data on a healthy-looking pairing, make any edit, or toggle Read Mode off and back on.

KEYBOARD SHORTCUTS

1	Go to Projects tab
2	Go to Sprint History tab
3	Go to Forecast tab
4	Go to Settings tab
5	Go to About tab
?	Show / hide the Keyboard Shortcuts help overlay

■ Shortcuts are disabled while a text input, dropdown, or date field has focus. Press ? at any time from any tab to see the shortcut list inside the app.

DISTRIBUTION TYPES AT A GLANCE

T-Normal	Truncated Normal — symmetric bell curve restricted to non-negative velocities; classical PERT-style shape. A reasonable alternative when sprint variability is low and approximately symmetric — at high variability the lower-bound truncation biases the mean upward and makes forecasts artificially optimistic. Available in both History and Subjective modes.
Lognormal	Lognormal — right-skewed curve always above zero, with a long upper tail; the recommended default because it matches the empirical right-skew of sprint velocity and remains well-calibrated even when sprint-to-sprint variation is large. **Enabled by default;** other distributions require opt-in via Settings → Statistical methods to show. Available in both History and Subjective modes.
Gamma	Gamma — right-skewed like Lognormal but with a thinner upper tail. Useful when faster sprints happen but extreme breakouts are unlikely. Adjusts more aggressively to velocity outliers than Lognormal.
Bootstrap	Bootstrap — resamples directly from your actual sprint history; the most data-driven option, assuming only that future sprints will look like past ones. Requires 5+ included sprints (History mode only). Most reliable with 10+ representative sprints.
Triangular	Triangular — a simple peaked shape with hard limits at ±3 standard deviations from the mean. Useful when you want a transparent, bounded forecast without long tails. Available in both History and Subjective modes.
Uniform	Uniform — every velocity in the range equally likely; the most conservative shape. Useful when you have little basis to prefer any one velocity value. Subjective mode only.

■ When multiple distributions are enabled and agree on a percentile date, that consensus signals a robust forecast. Significant divergence indicates sparse or volatile velocity data — consider enabling additional distributions in Settings → Statistical methods to show, and gathering more sprint history before committing to a deadline.

■ Set a lower trial count (1,000–5,000) for fast exploratory forecasting, then increase to 10,000–50,000 before a stakeholder presentation for smoother, more stable percentile results.

FORECAST GUIDANCE FOR NEW USERS

Choosing a percentile	Use **P50** for internal team targets — equal odds of finishing early or late. Use **P80** for stakeholder commitments — 4 in 5 chance of on-time or better delivery; this is the most common choice for sprint-based Agile releases. Use **P90–P95** for contracts, hard external deadlines, or releases with downstream dependencies. Avoid anchoring on P99 — it adds excessive buffer and rarely improves stakeholder trust.
Sprint count thresholds	**0–1 sprints:** Subjective mode only — History mode requires 2+ included sprints. **2–4 sprints:** History mode available; T-Normal, Lognormal, Gamma, and Triangular distributions active; an amber warning notes that forecast spread may be understated. **5+ sprints:** Bootstrap distribution unlocks, replacing Uniform. **10+ sprints:** Distributions typically converge — most reliable results. The more representative sprint history you have, the narrower and more trustworthy the forecast spread.
Forecast date precision	Projections show **sprint finish dates**, not specific days within a sprint. A productivity adjustment that reduces velocity by a small amount may not shift the projected end date if the adjusted velocity still completes the remaining work within the same sprint — this is correct behavior, not a bug. To model a full sprint gap (e.g., a team fully offline for a sprint), set the productivity factor to 0% for that date range, or extend the affected sprint using a custom finish date.

Backlog field units	The *Remaining Backlog* field on the Forecast tab and the *Backlog at End* field on the Sprint History tab must be entered in the same unit of measure configured for the project — the same unit used to record sprint velocity. Example: if the project is set up in story points and the team has 41 backlog items worth roughly 1,100 story points, enter **1,100** — not 41. Entering the raw item count when the project unit is story points will dramatically underestimate scope and produce an optimistic but incorrect forecast. If the project unit is *items*, then entering the item count is correct.
Reading chart signals	**Histogram:** a narrow peak = consistent team velocity and a tight forecast; a wide spread = high variability; right skew = occasional high-output sprints pulling the mean upward. **CDF:** a steep S-curve = low variance and a tight forecast; a gradual slope = wide uncertainty; a nearly vertical step with nearly identical percentile dates suggests all sprints had the same velocity or you have too few data points — add more sprint history before committing to a deadline.

■ If all distribution columns in the percentile table show the same completion date, you likely have identical sprint velocities (std dev = 0) or fewer than 3 sprints of history. Switch to Subjective mode and select a CV that reflects your team's expected variability, or add more sprint history before making stakeholder commitments.